

Trylto: Discrete IMEX Cloth Simulation with Learned Residual Forces

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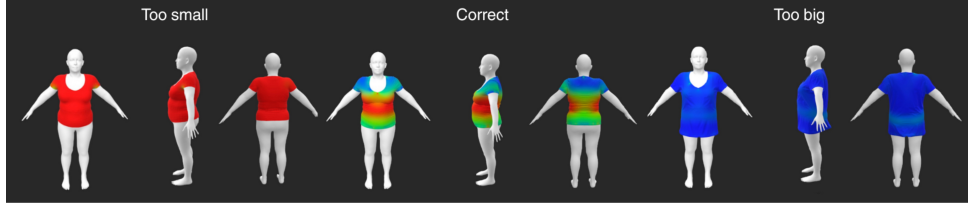


Figure 1: Strain distribution across garment sizes. Computed with the Green–Lagrangian tensor under the Kirchhoff–Love thin-shell formulation, revealing high-tension (red) and relaxed-drape (blue) regions across three size variants (blue = low strain, red = high)

Abstract

We propose *Trylto*, a hybrid cloth simulation framework that combines Kirchhoff–Love thin-shell mechanics with learned residual forces. The method introduces a time-varying (4D) latent UV atlas to improve texture-space continuity under large deformations, and a linearised IMEX time integrator that treats stiff elastic and damping terms implicitly whilst updating residual and external terms explicitly. To align optimisation with deployment, the framework minimises a discrete residual that mirrors the deployed numerical scheme, removing train–test drift typical of continuous-time PINN residuals under stiff time discretisations. We focus on the mathematical formulation, stability and conservation analysis, and computational complexity, positioning Trylto as a physics-informed foundation for high-fidelity, real-time garment simulation.

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1 Introduction

Cloth simulation faces a long-standing trade-off: classical finite element methods offer high accuracy but can be computationally prohibitive, whilst purely data-driven approaches enable fast inference but lack physical guarantees. We present *Trylto*, a hybrid framework that integrates analytical Lagrangian thin-shell mechanics with learned residual forces. A central design choice is *training–deployment consistency*: rather than optimising against

a continuous-time loss, we use a discrete residual that matches the deployed time integrator. This ensures the learned model is optimised in the numerical context in which it is deployed, avoiding integration drift. A time-evolving 4D latent UV atlas further maintains texture-space coherence under extreme deformations, improving continuity and reducing drift.

2 Physics-Informed Hybrid Framework

We represent cloth as a differentiable mapping $x(u, v, t) : D \times \mathbb{R} \rightarrow \mathbb{R}^3$. Let $s_i(t) = [x_i(t), \dot{x}_i(t)]^\top$ denote the state of vertex i . The semi-discrete dynamics are

$$\frac{d}{dt} \begin{bmatrix} x_i \\ v_i \end{bmatrix} = \begin{bmatrix} v_i \\ m_i^{-1} (F_{\text{phys}}(x) + F_{\text{ext}}(x, t) + F_{\text{contact}}(x) - C v + f_\theta(x, v, t)) \end{bmatrix}_i \quad (1)$$

2.1 Training–deployment consistency

We replace the continuous-time PINN loss with the discrete residual of the deployed IMEX step:

$$\mathcal{L}_{\text{IMEX}} = \sum_n \left\| M \frac{v^{n+1} - v^n}{\Delta t} - F_{\text{imp}}(x^{n+1}, v^{n+1}) - F_{\text{exp}}(x^n, v^n, t^n) \right\|_2^2 \quad (2)$$

This alignment removes train–test drift typical of continuous-time PINN residuals under stiff discretisations.

2.2 Dynamic UV atlas

Each vertex carries latent coordinates $z_i(t) \in \mathbb{R}^4$, learned jointly with network parameters and updated by

$$z_i^{n+1} = z_i^n - \eta_z \nabla_{z_i} \left[\sum_{(i,j)} \|z_i - z_j\|^2 + \beta \|z_i^n - z_i^{n-1}\|^2 \right] \quad (3)$$

Why 4D? In three dimensions, any continuous parameterisation of a surface necessarily introduces singularities (cf. the hairy ball theorem [Milnor 1997]). By embedding in 4D spacetime coordinates

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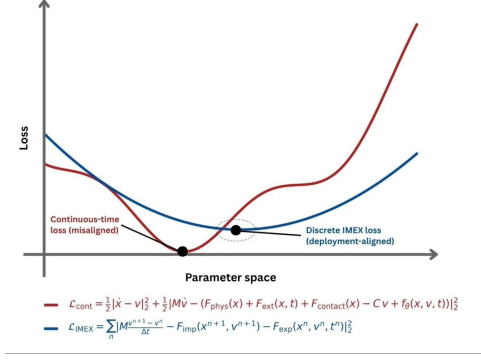


Figure 2: The continuous-time objective (red) converges to a misaligned optimum, while the discrete IMEX formulation (blue) aligns training with the deployed solver.

$z_i(t) = [u, v, w, T]$, where T represents a temporal dimension, we provide sufficient degrees of freedom to displace singularities while maintaining temporal coherence. This spacetime embedding allows the atlas to evolve smoothly through deformation history, avoiding the topological constraints that plague purely spatial parameterisations.

3 Mathematical Foundations

We adopt a Kirchhoff–Love thin-shell model [Grinspun et al. 2003] with Lagrangian $L = T - U$,

$$T = \frac{1}{2} \int_D \rho \|\dot{x}\|^2 dA, \quad U = \int_D W(F, n) dA, \quad (4)$$

and an anisotropic textile energy

$$W(F) = \frac{1}{2} k_a (\lambda_a - 1)^2 + \frac{1}{2} k_b (\lambda_b - 1)^2 + k_{ab} E_{ab}^2 + W_{\text{bend}}, \quad (5)$$

where λ_a, λ_b are stretch ratios along warp/weft directions and W_{bend} models curvature-dependent bending.

3.1 Strain visualisation

We evaluate strain using the *Green–Lagrangian* tensor, appropriate for large, anisotropic deformations consistent with the thin-shell model. The resulting strain maps directly reveal anisotropic tension and relaxation patterns, validating how the analytic model captures fabric–body interaction (Figure 1).

4 Production-Grade IMEX Integration

4.1 Force split

$$F_{\text{imp}}(x, v) = F_{\text{phys}}(x) + F_{\text{coll}}(x) - C v, \quad F_{\text{exp}}(x, v, t) = f_{\theta}(x, v, t) + F_{\text{ext}}(x, t). \quad (6)$$

4.2 Linearised IMEX step

With timestep Δt ,

$$x^{n+1} = x^n + \Delta t v^{n+1}, \quad M \frac{v^{n+1} - v^n}{\Delta t} = F_{\text{imp}}(x^{n+1}, v^{n+1}) + F_{\text{exp}}(x^n, v^n, t^n). \quad (7)$$

Linearising F_{imp} at (x^n, v^n) and writing $\Delta x = x^{n+1} - x^n$ gives

$$A \Delta x = b, \quad (8)$$

$$A = \frac{M}{\Delta t^2} - \left(\frac{\partial F_{\text{phys}}}{\partial x} + \frac{\partial F_{\text{coll}}}{\partial x} + \frac{\partial(-Cv)}{\partial x} \right) \bigg|_{x^n} + \frac{C}{\Delta t}, \quad (9)$$

$$b = \frac{M}{\Delta t} v^n + F_{\text{phys}}(x^n) + F_{\text{coll}}(x^n) - C v^n + F_{\text{exp}}(x^n, v^n, t^n). \quad (10)$$

4.3 Stability

The implicit block is A-stable for the linearised stiff modes (elasticity and damping), while the explicit block obeys

$$\Delta t \leq \frac{2}{\sqrt{\lambda_{\max}(M^{-1}K_{\text{exp}})}}, \quad K_{\text{exp}} = \frac{\partial F_{\text{exp}}}{\partial x} \bigg|_{(x^n, v^n, t^n)}. \quad (11)$$

5 Theoretical Properties

5.1 Conservation

In the conservative limit (no damping/contact) and with internal forces derived from a potential, linear and angular momentum are preserved up to solver tolerance; energy behaviour is bounded by the IMEX discretisation.

5.2 Convergence

Under the assumption that the analytic model captures the dominant elastic and inertial response, the discrete residual minimisation drives $f_{\theta} \rightarrow 0$, with convergence rate dependent on the conditioning of the IMEX system matrix.

5.3 Stability

The hybrid scheme inherits A-stability from the implicit treatment of stiff modes, while the explicit block remains subject to a CFL-type condition.

5.4 Computational efficiency

Assembly and solve are superlinear but sub-cubic in N (e.g., $O(N^{1.2} - N^{1.5})$) in practice with modern sparse solvers, significantly cheaper than fully implicit $O(N^3)$ methods, and amenable to GPU parallelisation.

6 Conclusion

We have outlined a hybrid cloth formulation combining Kirchhoff–Love thin-shell mechanics, a 4D time-evolving UV atlas, and a linearised IMEX integrator supervised by its own discrete residual. The formulation unites the stability of established numerical methods with the flexibility of learned refinements, enabling applications such as interactive virtual try-on, garment retargeting, and physically grounded design workflows.

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